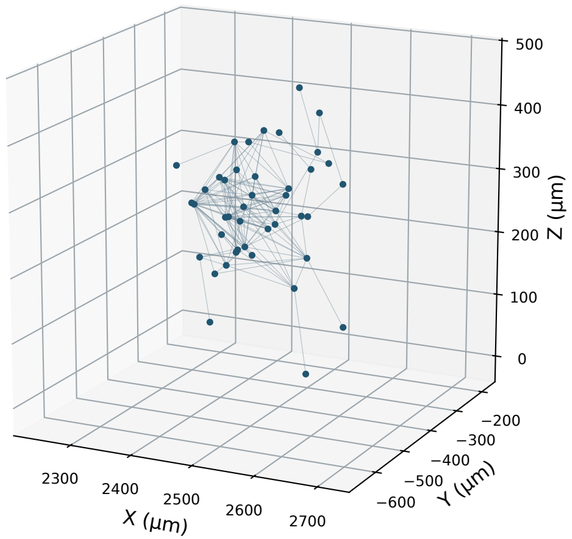
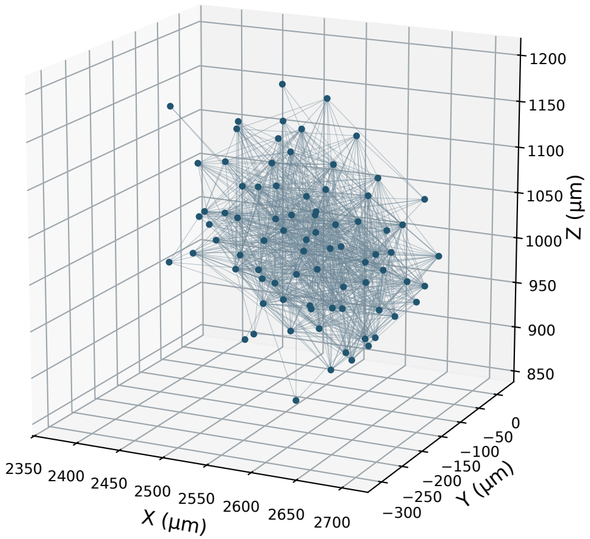
FIG 01 | Why connected fragments lost the old surface-load representation

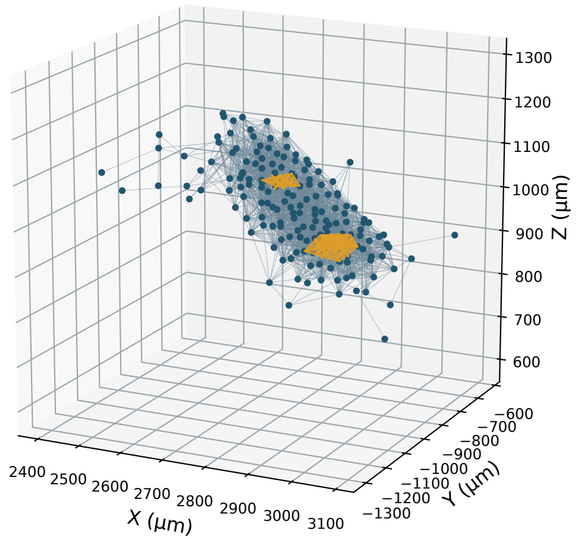
Uniform ID 1164: 43 nodes
0 supported tets; 0 loaded facets



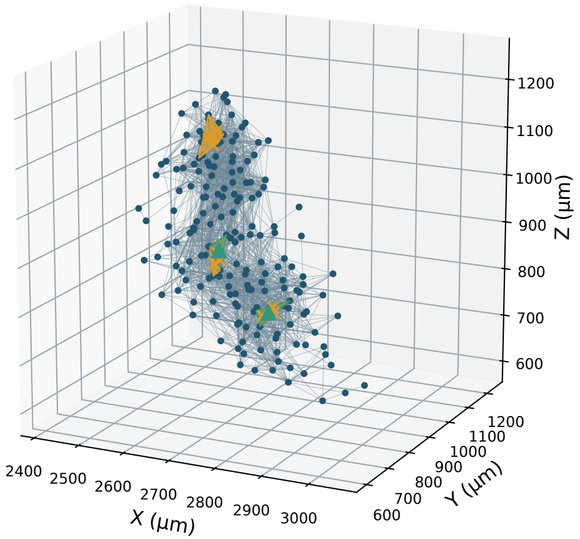
Uniform ID 1804: 81 nodes
0 supported tets; 0 loaded facets



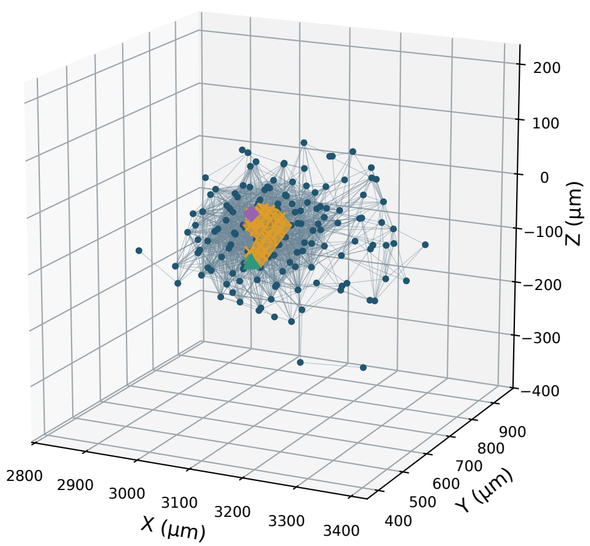
Uniform ID 1848: 167 nodes
5 supported tets; 0 loaded facets



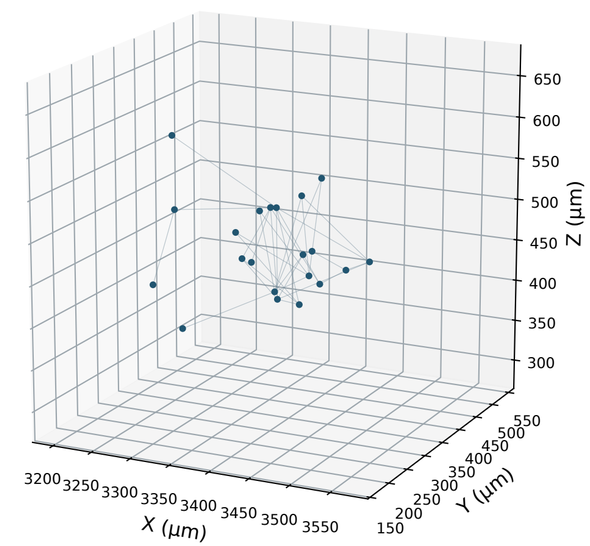
Uniform ID 1884: 181 nodes
3 supported tets; 2 loaded facets



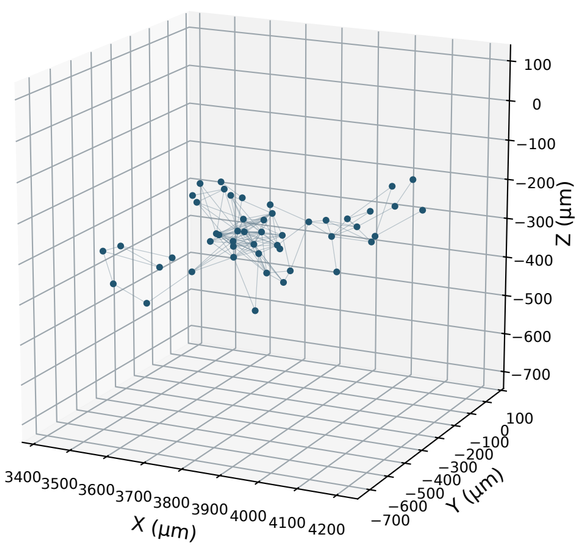
Biphasic ID 3769: 148 nodes
12 supported tets; 1 loaded facets



Biphasic ID 3809: 21 nodes
0 supported tets; 0 loaded facets



Biphasic ID 3846: 49 nodes
0 supported tets; 0 loaded facets



Actual V2 end states

Blue: real material points
and surviving original bonds

Orange crosses: unopened candidates
Purple diamonds: open, inaccessible
Green triangles: loaded face centroids
(markers are identification symbols;
overlaid for visibility, no position shift)

4 fragments: no supported tetra
1 zero-load fragment: opening test fails
2 others: opening/accessibility limits

No geometric shell was invented.
No loaded facet ≠ no physical drag.
Original world coordinates throughout.